
Project: NLP-based Customer Review Analysis System

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1. Introduction

This report outlines the development of an automated system that analyzes customer product reviews using NLP models. The system consists of three main tasks: sentiment classification, product category clustering, and summarization of reviews into concise, actionable insights. Given the ever-growing volume of customer feedback on e-commerce platforms, manually analyzing reviews is no longer efficient. The system uses advanced NLP models such as BERT for classification, K-Means for clustering, and BART for generative summarization. The solution also integrates with Gradio to provide an interactive web interface for users.

2. Review Classification

Objective:

To classify customer reviews into three sentiment categories: positive, neutral, and negative, in order to aid companies in improving their products and services.

Task:

We used a model that analyzes the textual content of customer reviews and assigns them to one of the three sentiment categories.

Dataset Description:

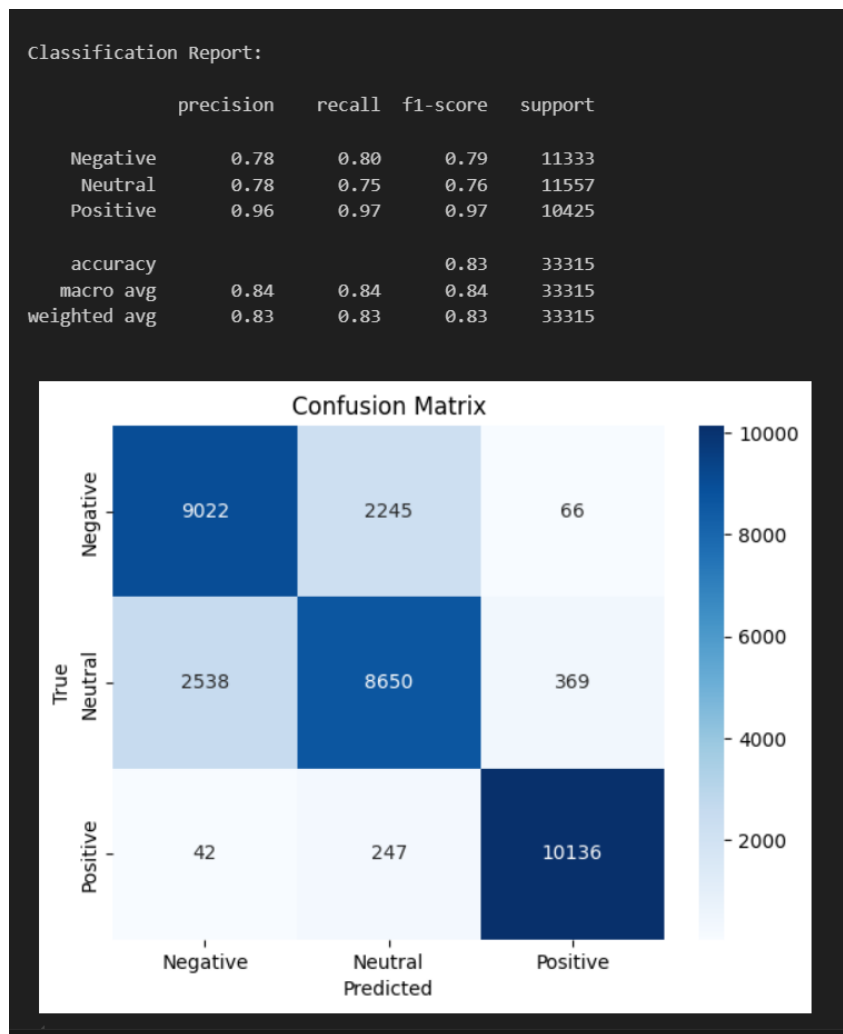
The dataset used is the 'Amazon Customer Reviews' dataset, which contains over 20,000 reviews. The reviews are categorized based on star ratings, mapped to three sentiment labels:

- Positive: Rating 4-5
- Neutral: Rating 3
- Negative: Rating 1-2

Model Development:

1. **Importing Libraries**: We use libraries such as `transformers` and `torch` for model development and tokenization.
2. **Loading the Dataset**: We load CSV files containing the reviews and their sentiment labels.
3. **Label Encoding**: The sentiment classes are converted into numerical labels for compatibility with the model.
4. **Tokenization**: Review text is tokenized using `BertTokenizer` to prepare it for the BERT model.

5. **Model Training**: A pretrained **BERT model** is fine-tuned using the `Trainer` API with adjusted class weights.`
6. **Model Evaluation**: We evaluate the model's performance using precision, recall, and F1-score.



3. Product Category Clustering

Objective:

To cluster customer reviews into broader categories (e.g., Electronics, Household Supplies) for easier analysis.

Task:

Develop a model that groups customer reviews into 4-6 high-level categories based on product features.

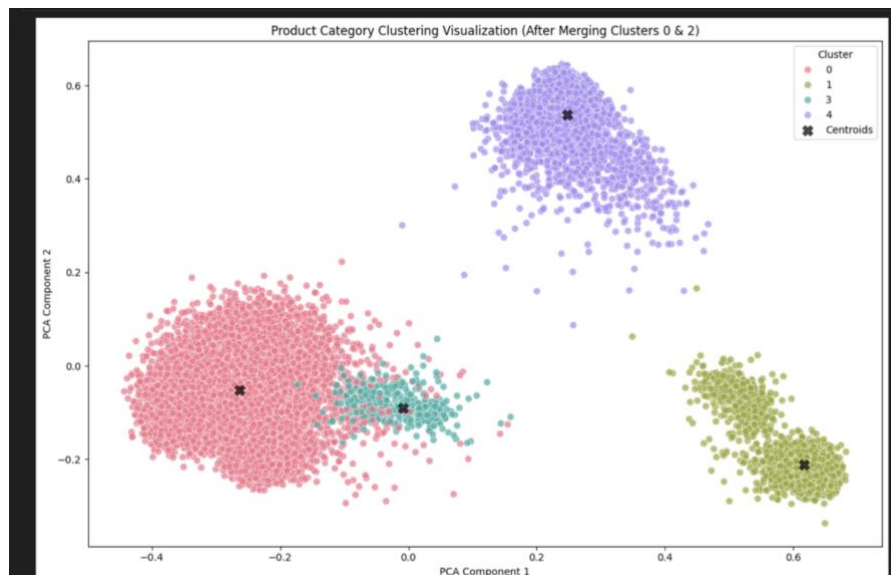
Dataset Description:

We use a dataset with over 34,000 reviews across various product categories. The dataset includes columns such as:

- `product\_title`
- `reviews.text`
- `cluster\_label`

Model Development:

1. **Libraries**: We utilize SentenceTransformer and KMeans clustering to generate embeddings for reviews.
2. **Embedding Generation**: Reviews are converted into embeddings using the **SentenceTransformer**.
3. **Clustering**: We apply **KMeans** to cluster the product reviews based on their embeddings.
4. **Balancing the Dataset**: We use undersampling to ensure that popular product categories don't dominate the clustering process.
5. **Visualization**: using scatter plot



4. Review Summarization Using Generative AI

Model Development:

We used **BART** (Bidirectional and Auto-Regressive Transformers) for text summarization. The model generates human-like summaries that present a story-like view of the product category based on user reviews.

Deployment:

1. **API Endpoint**: We provide an API endpoint for generating summaries (`/summarize`).
2. **Gradio Interface**: The system uses **Gradio** to allow users to select a product category and generate a review summary.

The screenshot shows the 'Product Review Analysis App' interface with the 'Category Clustering' tab selected. It features a text input field for 'Enter product name or review' containing the word 'tablets'. Below the input is a label 'Predicted Category: Electronics'. At the bottom of the main content area is a button labeled 'Predict Cluster'. The footer contains the text 'Use via API', 'Built with Gradio', and 'Settings'.

The screenshot shows the 'Product Review Analysis App' interface with the 'Summarization' tab selected. It features a dropdown menu for 'Select Cluster Category' with 'Electronics' selected. Below the dropdown is a text area for the 'Summary' containing a paragraph of text about Kindle Paperwhite, Fire TV Stick, and Amazon Fire TV. At the bottom of the main content area is a button labeled 'Generate Summary'. The footer contains the text 'Use via API', 'Built with Gradio', and 'Settings'.

5. Limitations and Challenges

- **Data quality and imbalance**: The dataset has a disproportionate number of positive reviews, which makes training challenging.
- **Clustering Complexity**: Deciding on the right clustering method and features to use was a challenging task.
- **Model Coordination**: Coordinating the outputs from classification, clustering, and summarization models proved to be complex.

6. Conclusion

This report presents an end-to-end NLP-based solution for automating the analysis of customer product reviews. The system combines sentiment classification, product clustering, and AI-generated summarization to offer valuable insights from customer feedback. By deploying these models in a user-friendly web interface using **Gradio**, the system allows for real-time interaction, making it an effective tool for businesses to understand and improve their products.